Question 1 *Which of the following is not in our integer sandbox? In other words, what can't we assume in COMP61?*

Multiple Choice:

(a) *Integers*

(b) *Addition*

(c) *Subtraction*

(d) *Multiplication*

(e) *Division*

(f) *Inequalities*

Question 2 *Fill in the missing part of the definition from the options given: Definition: an integer is called* **even** *provided it* ______.

Multiple Choice:

(a) *ends with 2, 4, 6 or 8.*

(b) *is not above 100*

(c) *can be written as* $2x$ *for some number* x

(d) *is divisible by 2*

Question 3 *Fill in the missing parts of the definition from the options given: Divisible: a is divisible by b if there exist a/an* ______ *such that* ______.

Multiple Choice:

(a) *number c;* $c = b$

(b) *number c; b times c does not equal a*

(c) *integer c; b times c equals a*

(d) *number c; b times c equals a*

Question 4 *Fill in the missing parts of the definition from the options given: Odd: An integer a is called odd provided there is ______ such that ______.*

Multiple Choice:

(a) *a rational number x; a = 2x*

(b) *an integer x; a = 2x + 1*

(c) *an integer x; a = x*

(d) *any number x; a = 2x + 1*

Question 5 *Fill in the missing parts of the definition from the options given: Prime: x is prime provided that ______ and the ______ of x is/are ______.*

Multiple Choice:

(a) *x > 0; only positive divisor; 1*

(b) *x > 0; only divisor; 1*

(c) *x > 1; only divisors; 1 and x*

(d) *x > 1; only positive divisors; 1 and x*

Question 6 *What are the correct ways to say a is divisible by b?*

Select All Correct Answers:

(a) *b divides a*

(b) *b is a factor of a*

(c) *a is a divisor of b*

(d) *b is a divisor of a*

(e) *b | a*